Problem Statement and Goals
Software Engineering

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Table 1: Revision History

Date	Developer(s)	Change
Sep 27, 2026	All	Added first draft (Inputs/Outputs, Goals)
Sep 28, 2026	All	Added remaining sections
Sep 28, 2026	All	Added Reflection

1 Problem Statement

1.1 Problem

Currently, speech-language pathologists (SLPs) evaluating bilingual children through telehealth face challenges when annotating and managing their patient assessments. Addressing these limitations is important for supporting accurate and efficient patient assessment, particularly in pediatric telehealth settings where clinicians and patients rely on digital tools. The existing annotation tool, SAFA, is hosted locally and operates separately from the telehealth web platform used by patients and their guardians. In addition, SAFA has a limited and unintuitive user interface, which contributes to a poor user experience for clinicians. Its functionality also does not fully support the clinician-side assessment, record-keeping, and analysis workflow required by SLPs. In summary, there is a need for better system integration between assessment tools and telehealth workflows, improved usability for clinicians, and more comprehensive functionality for recording and analyzing patient assessments.

1.2 Inputs and Outputs

Inputs: A video recording assessment of a bilingual child and speech clinician completed on a web-based user-centered platform.
Outputs: A web-based annotation interface, integrated directly into the TeleHealth platform, that allows a speech clinician to review, edit, and export annotated transcriptions of the input video.

1.3 Stakeholders

1.3.1 Primary Stakeholders

Speech-language pathologists (SLPs): SLPs, or speech clinicians, are the direct users of the integrated system. They review assessment recordings, verify and correct AI-generated transcripts, tag speakers and languages, and use the results to make diagnostic decisions and write reports. Their main needs are less time spent annotating, an interface that requires little training, and full control over the final decision.

1.3.2 Secondary Stakeholders

Children and guardians: Children complete assessments on the patient-facing site and are supervised by their guardians. They do not use our system directly, but its quality affects how quickly and accurately their assessments are reviewed and how soon they receive support.

Research team (Dr. Du, Dr. Wu, and collaborators): They developed SAFA and the surrounding research, and they want the integrated system to become part of a larger clinician-support platform. They depend on our work meeting clinician requirements and being compatible with their existing pipeline.

1.3.3 Tertiary Stakeholders

Clinics, schools, and healthcare institutions: Organizations that employ SLPs must approve tools that handle patient data and may be affected by changes to their clinical workflows.

Privacy and regulatory bodies: Because the system stores recordings and records of children, it must respect applicable health-privacy requirements (e.g., HIPAA for U.S.-based clinicians).

1.4 Environment

1.4.1 Software

The system will be a web-based application accessed through the existing TeleHealth platform, and will be supported on all current web browsers. The platform itself will be built using modern web frameworks such as React, and languages like Python. Modules will follow common software design patterns and

object-oriented programming principles. Additional technologies and libraries will also be used for purposes such as cloud services in order to host the platform, data storage, and authentication.

1.4.2 Hardware

Clinicians will access the annotation interface from a desktop view, as reviewing and editing transcripts alongside video playback requires a larger screen, as well as a keyboard and mouse for precise editing. Audio output, such as speakers or headphones, will also be required to review speech and videos from recorded assessments. Patients and caregivers will complete assessment sessions through TeleHealth's existing web-based interface, using a device with a webcam and microphone. Support for assessments and annotations on mobile phones or tablets will be a stretch goal but may likely be out of scope for this project.

2 Goals

1. **Integrate SAFA Into a Telehealth Platform**: SAFA is currently a local, standalone tool, requiring clinicians to leave the TeleHealth platform entirely to annotate assessments. Success will be measured by clinicians being able to access and use SAFA's core annotation features entirely within their TeleHealth account, with no local installation or file transfer required.

2. **Improve SAFA's Usability**: Currently SAFA is not very intuitive which may discourage many users. We aim to redesign the annotation interface to be simpler and more guided, so that any speech clinician can use the software seamlessly. Success will be measured through usability testing (task completion rate, or a post-use satisfaction survey) showing improved ease of use compared to the current SAFA interface.

3. **Speed Up Speech Annotation**: The integration of SAFA into the TeleHealth platform will speed up the annotation process for bilingual speech assessments conducted through the platform. Manual annotation of a single session can take up to 3-4 hours whereas using SAFA has already been shown to cut this by roughly one-third. We aim to preserve or improve that amount of time once integrated into the TeleHealth platform. It will be measured by comparing average annotation completion time before and after integration.

4. **Integrate a Clinician Record and Analysis Hub**: Centralizing all patient records and assessment history in one accessible hub will simplify clinicians' workflow, allowing them to quickly reference patient history without navigating multiple disconnected systems. Success will be measured by clinicians retrieving a specific past assessment in under a set number of clicks or seconds, then comparing them to the current process of locating the files in a distributed system.

3 Stretch Goals

1. **Provide Patient-Facing Annotation Reports**: Allow clinicians to generate patient assessment reports containing relevant annotations and clinician notes. These reports would provide patients and their guardians with a clear summary of assessment results and support their engagement with the ongoing treatment process. Success may be measured by the successful generation of a complete and accurate summarized assessment in a shareable format, such as a PDF.
2. **Support Multiple Assessment Comparison**: Allow clinicians to compare multiple assessment sessions for the same patient over time. This would support clinicians in tracking changes in pediatric patients across assessments and identifying relevant trends in patient performance. Success may be measured by clinicians being able to view multiple assessments side-by-side and access a visualization of the patient's assessment history.

4 Custom Documents (CDs)

The custom documents chosen for the project are:

1. **Wireframe (Fall)**: Since redesigning SAFA's interface is a core goal, we will create a mock-up of the new interface and present it to the research team and clinicians. The document will describe the mock-up and discuss insights gained from their feedback.
2. **Usability Report (Winter)**: Clinicians should be able to use the integrated system with ease and have it suit their daily workflows. This document will include the testing plan, the results, and a discussion of the findings and user feedback (including improvements on how the system looks and functions).

Appendix — Reflection

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

Yilin

This deliverable, I think the planning and task load delegation were most important to the final product. Despite preferring in-person meetings, we had an online meeting prior to beginning drafting this deliverable. In the meeting, we assigned each team member to different parts based on workload, interest, and expertise. This was especially important this deliverable as the team only received critical project information and requirements late into the timeline. Under time constraints, having an immediate meeting to ensure we were all on the same page and knew our next steps was important for maximizing writing time and quality.

While writing the problem statement, I struggled with separating the clinician's (our primary stakeholder) "why" from the more technical point of view our supervisor's team had reported. I worried that I would misrepresent the clinician's perspective, as I only had direct communication with the system designers. For instance, though both groups would benefit from a more user-centered interface, the problems causing that need differ (where the clinician would have real-time patient use case difficulties). To resolve this issue, I focused on understanding different clinician's points of view in the research paper draft our supervisor shared with us. Another pain point was that when writing the stretch goals, I found it difficult to ensure they were clearly second in priority to our main goals. To resolve this issue I compared the stretch goals and primary goals

against the three main task descriptions our supervisor provided for us. The stretch goals were verified to be both extended from the initial project scope while still remaining relevant and impactful if fulfilled.

Lastly, the scope of the goals was also derived from the three main task descriptions our supervisor provided. In a meeting with our supervisor, they ensured the tasks were suitable for a Capstone project based on their prior years of experience working with Capstone teams. Referencing the task list and our problem statement helped us to keep within the stakeholder needs. With these tasks as our baseline, we ensured all goals were measurable. This meant that each goal would be accompanied by a success metric that our team could feasible measure and fulfill.

Catherine

Our team was given a wide breadth of resources and knowledge from both our supervisor and additional professors also working on the MERS project, so we were able to define our problem and goals pretty well before starting this deliverable. We also planned multiple meetings in order to discuss each task item, brainstorm various ideas/notes, and ensure that everybody was on the same page before we broke off to individually tackle our assigned parts. Having both the project knowledge and pre-work discussions helped heavily in guiding us through our deliverable, so we were able to easily communicate our individual thoughts with each other and work through the deliverable efficiently.

One of the pain points I believe we experienced was understanding the nitty-gritty details of our project, given how it touches several moving parts, from the SAFA system to the TeleHealth platform. There were many questions we had about the project, and we often had to wait for responses, gather information from lengthy papers, and make our own assumptions in the meantime. What helped a lot was, again, our frequent and productive meetings, where we brainstormed and worked on understanding the project together. Once we had a better grasp of the project, another pain point was then figuring out our scope; e.g what can we realistically finish and do we need to scope up or scope down? To work through this, we outlined our main goals and separated those from our stretch goals, and plan on using our POC as a checkpoint since it validates our most concerning risks.

To keep our scope as realistic but properly complex as possible, we made sure that as a team, we fully and thoroughly understood the project and what was being asked of us. This included understanding what parts of the MERS system were done, what could be improved, and what needed to still be implemented or connected. By separating those into individual goals as previously mentioned, we were able to visualize and better understand the scope of our project and what could be doable within the next two terms. We wanted to ensure that we weren't underestimating any technical work involved, so actually working on this deliverable also brought up multiple useful questions that we asked ourselves;

such as what are the risks that come with your project and are we able to prevent them? Essentially, visualizing our project in a more in-depth manner, and breaking the tasks that we'd eventually work on developing into "modules" helped us a lot in determining what would be achievable this year.

Nour

What went well when writing this deliverable was brainstorming what to include in the problem statement. This is because our supervisor, Dr. Irene, made clear what she would like to add and change in the existing system. Additionally, she and her team provided several resources for us to understand the current system and its pain points. Moreover, since we knew what was expected of us and the priority level our tasks, we knew which Custom Documents would be best suited for the project. For example, we might write a Norman's Principles Report as one of our custom documents, considering one of our objectives is improving the usability of the SAFA system.

Although we had a good understanding of the SAFA annotation tool, we had hardly any understanding of the actual TeleHealth system it would be integrated into. This made it difficult to know what to include in the Expected Technology section of the Development Plan document. Especially considering myself, who is not experienced in web-development or audio-video synchronization tools. Therefore, pinpointing what tools the development team may have used to implement the TeleHealth system, and what we might need to use to integrate SAFA, was difficult.

The scope of the project was largely defined by our supervisor from the start, hence adjusting it was not much of an issue. When we first consulted with her, the tasks she outlined initially seemed lighter than what we expected from a capstone project. However, after clarifying this with her, she confirmed that the work was enough, with additional tasks available if we wanted to take on more. Hence, we broke down the main objectives she explicitly identified into our goals and added a few additional ones to account for possible extended work if time and resources allowed.

Javayria

This deliverable helped our team understand the project beyond the brief description we were given when it was pitched. What went best was writing the problem statement, because the supervisor, Dr. Irene Yuan, and the other researchers gave us notes on the current telehealth assessment system and SAFA. That made their baseline expectation clear from the start: integrate the systems into one larger application that speech-language pathologists can use for their work. I was also pleasantly surprised by how responsive and supportive the research team is, even though they are based in the U.S. We still don't know many details about the existing system or exactly what they are envisioning, but it's relieving to be working with a team that doesn't hesitate to help.

My main pain point was the GitHub management work. I am a big fan of structure and guidelines, so I wanted the team to use naming conventions and templates wherever possible. However, setting them up was much harder than expected as GitHub still doesn't offer a UI option to choose a template when creating a pull request. I originally wanted one template for documentation changes and another for code changes, but I ended up creating a single default template that is applied to every PR. Although this wasn't the focus of the deliverable, it reminded me that setting up CI/CD for code changes later on will likely be a challenge.

The scope has stayed mostly the same as the description Dr. Yuan gave us in our first supervisor meeting. The main change has been working out how to make it complex enough for a senior design project. At first, the goal was to integrate SAFA into the larger telehealth system. We were worried that this could either end up being a few adapters and some simple web development, or a full-fledged serverless architecture system that would take up most of our time. After learning more about SAFA and how Dr. Wu's lab implemented it, we found that there is a lot more work than making it available on the web. The tool is still in its early stages, and the research paper contains multiple feature requests from clinicians. The two systems will need to interact in several ways, and we will need to decide what additional components the combined system requires. I'm both excited and nervous, as the scope will need to be refined throughout the term once we learn more about the work we are inheriting.